

sible file to make sure that it remains easily portable. The package systems are developed around process collection of package files, and petrified is referred to as an archive file containing the compiled binaries and other resources.

The packages also provide convenience while storing and easily compress the files to limit their storage space. Binaries in packages are also precompiled according to the sane defaults chosen by the developer. Packages also input meta-data like software name, description of the purpose, the version number of the software, list of essential dependencies required by the software to run. Various instances of Linux have developed their package manager formats, and some of the highly concerned ones are-

- **.Deb:** this package format is utilised by Ubuntu, Debian, Linux mint and various other derivatives in the same segment.
- **.rpm:** this format was initially known as RedHat package manager, and it is utilized by RedHat, Fedora, SUSE, and various other distributions
- **.tar.xz:** the Arch Linux uses this package format, and it is just compressed tarball

Software repositories

Software repository refers to a centralised collection and listing of available software for the corresponding system or any other repository system which has been configured successfully to use. For example, for searching in a repository for any specific package, Arch Linux with Airman.

What are the popular package managers?

Package managers are used to interact with the software repositories. Let us go through some of the package managers -

- **RPM Package Manager (Red-Hat Package Manager)-** RPM Package Manager is a program for installing, uninstalling, and managing software packages in Linux. The “RPM” part of this term comes from .rpm, which is the default extension for files used by the program.
- **Debian based package manager-** It uses .Deb packages and can be easily managed by the tool called dpkg. It was designed to maintain the packages available locally effectively. This project package is commissioned with another package called Deity, to improve user experience.
- **Arch-based package managers-** arch Linux utilises a package manager known as Pacman and unlike .rpm or .deb, Pacman uses a traditional tarball with the help of LZMA2 compression (.tar.xz). This utility enables the packages to be smaller as compared to the other compressed archives.